Introduction to Cloud Computing (DV1566)

Project Report

Group 19

Naga Pujitha Pitta

Renuka Nandikolla

Santhosh Birada

❖ **Deployment of a multitier web application on the cloud, showing that the application is scalable. The deployment should foreseen the use of the cloud services (computation, storage, network management).**

The goal of the project is to build a multitier web application on the cloud and demonstrate its scalability by utilizing multiple cloud services for computing, storage, and network management. The deployment method will maximize resource consumption and efficiently accommodate shifting workloads. The use of cloud infrastructure enables flexibility, scalability, and cost-effectiveness, allowing the application to adapt to changing demands smoothly. The project will demonstrate how cloud services may improve the performance, stability, and scalability of web application architecture.

Explanation of the Design:

1. Initiate the setup by launching an EC2 instance.

2. Use the established EC2 instance to host the 'job-finding' website.

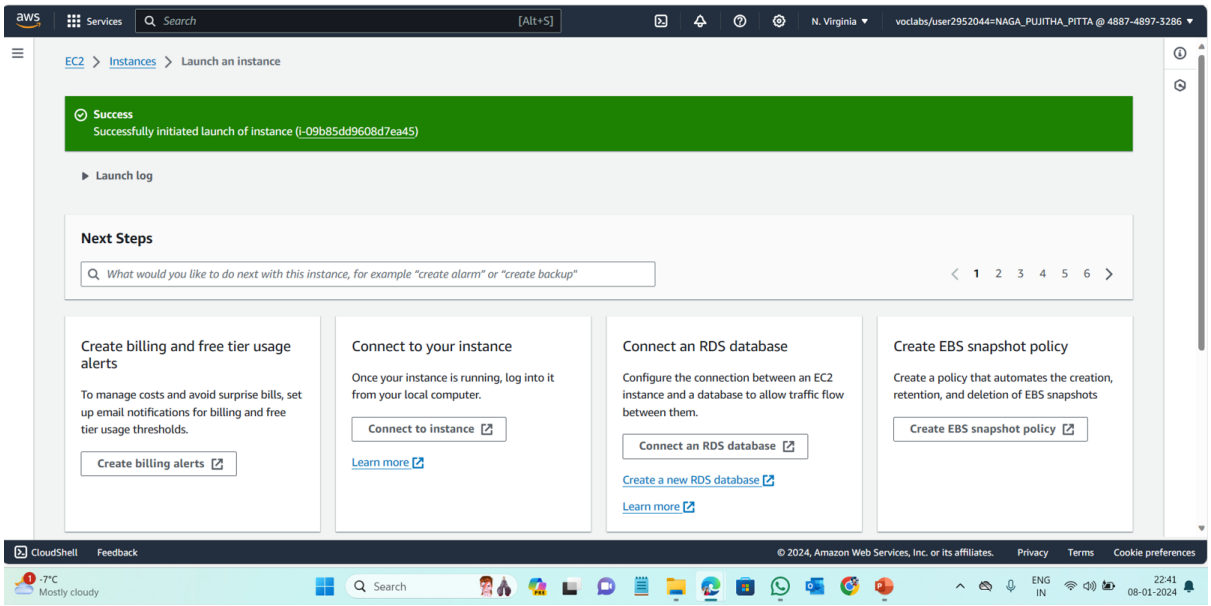
3. For replication, create an Amazon Machine Image (AMI) of the job-finding instance.

4. Create launch configuration and scaling groups to allow for dynamic scaling of the job-finding instance.

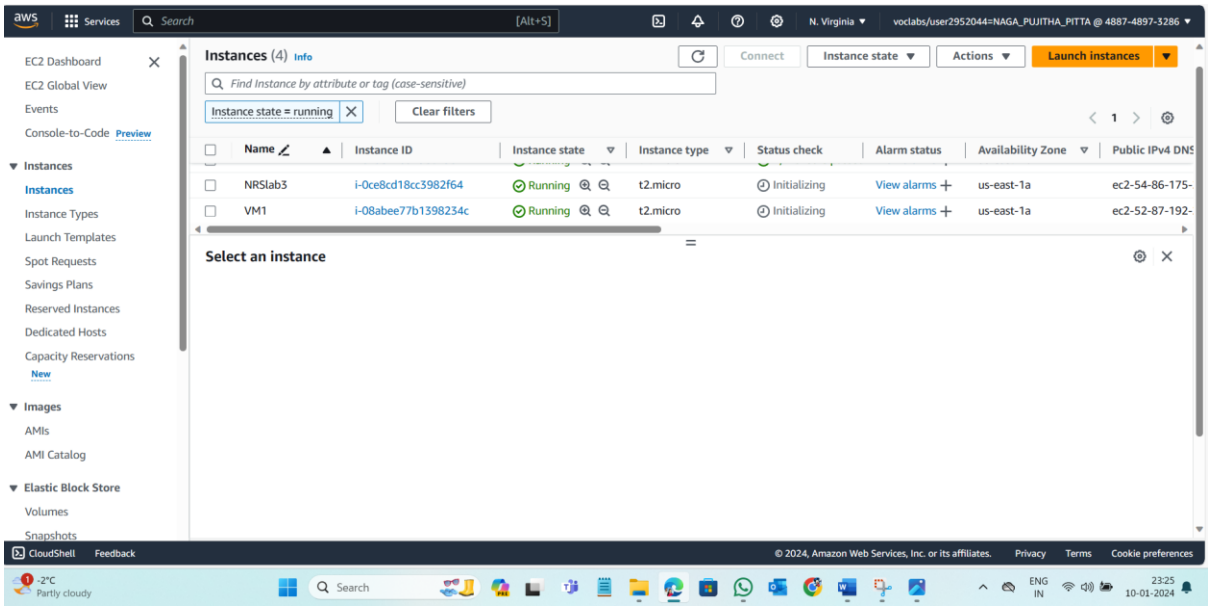
5. Use load balancing and target groups to improve traffic availability and distribution for the job-finding instance.

❖ Implementation:

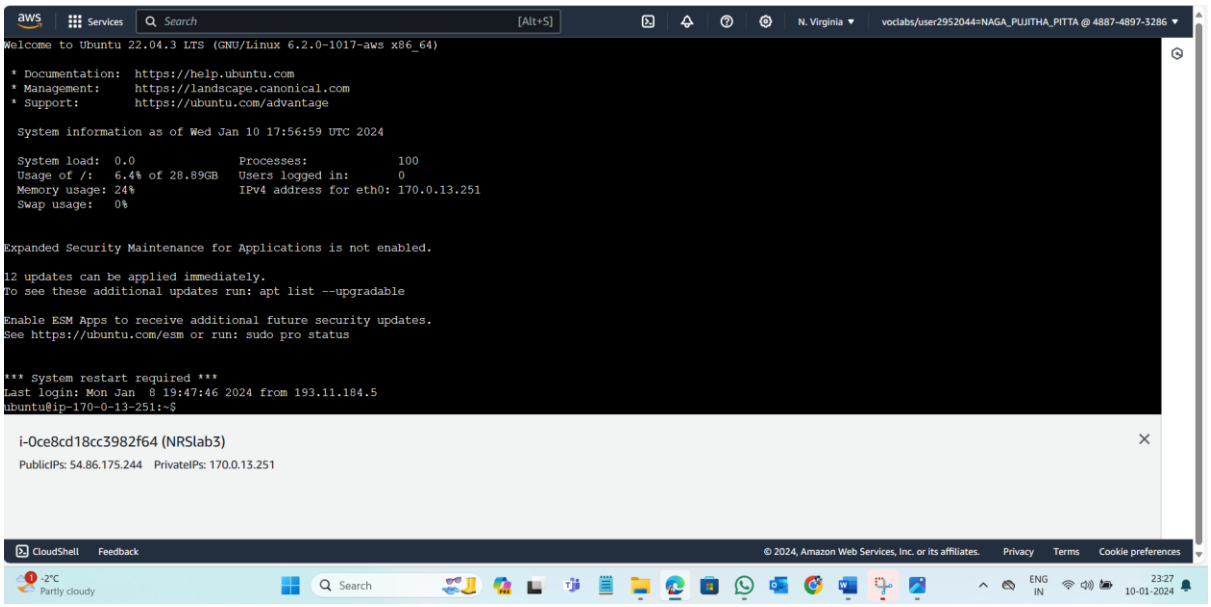
EC2 instance successfully created.



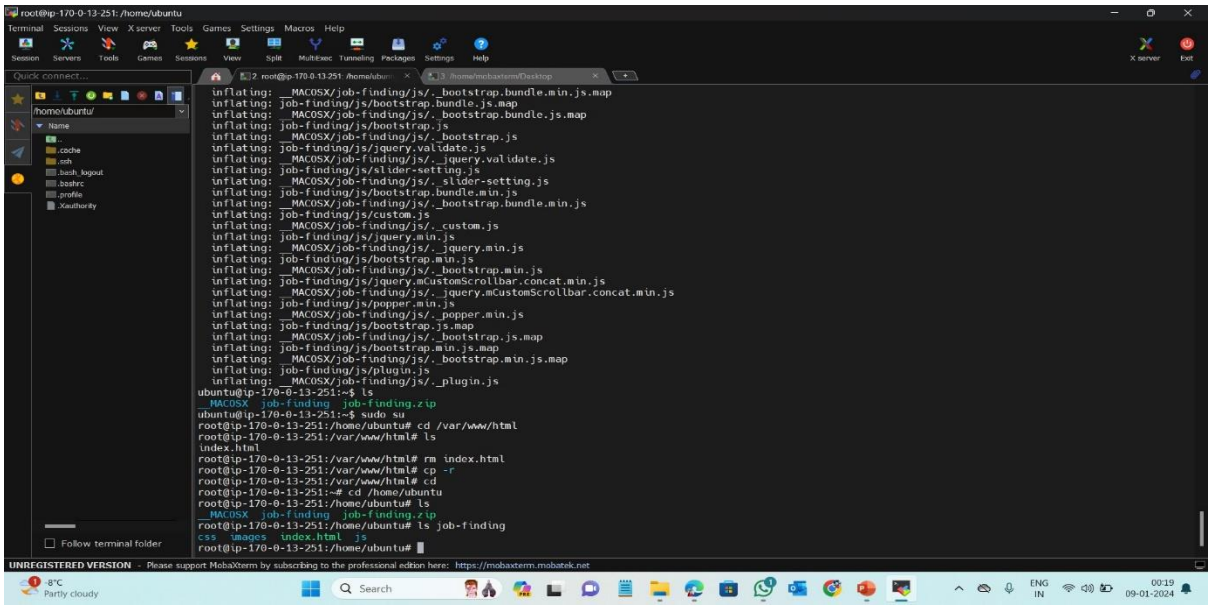
NRSlab3 has been successfully established, and it is now running in a successful condition.

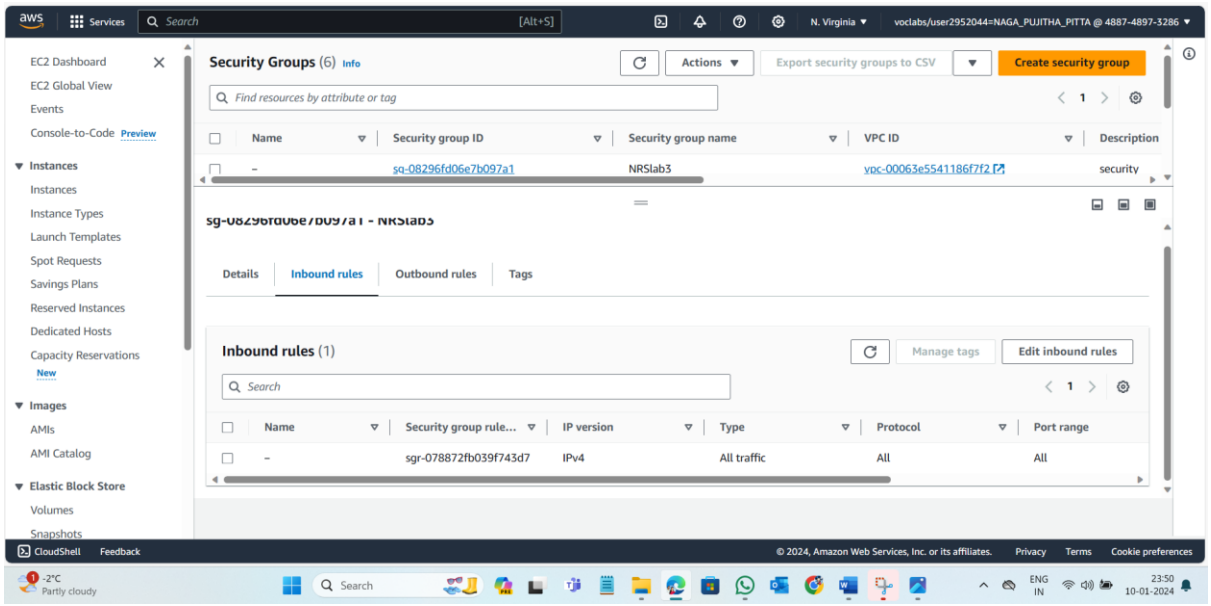


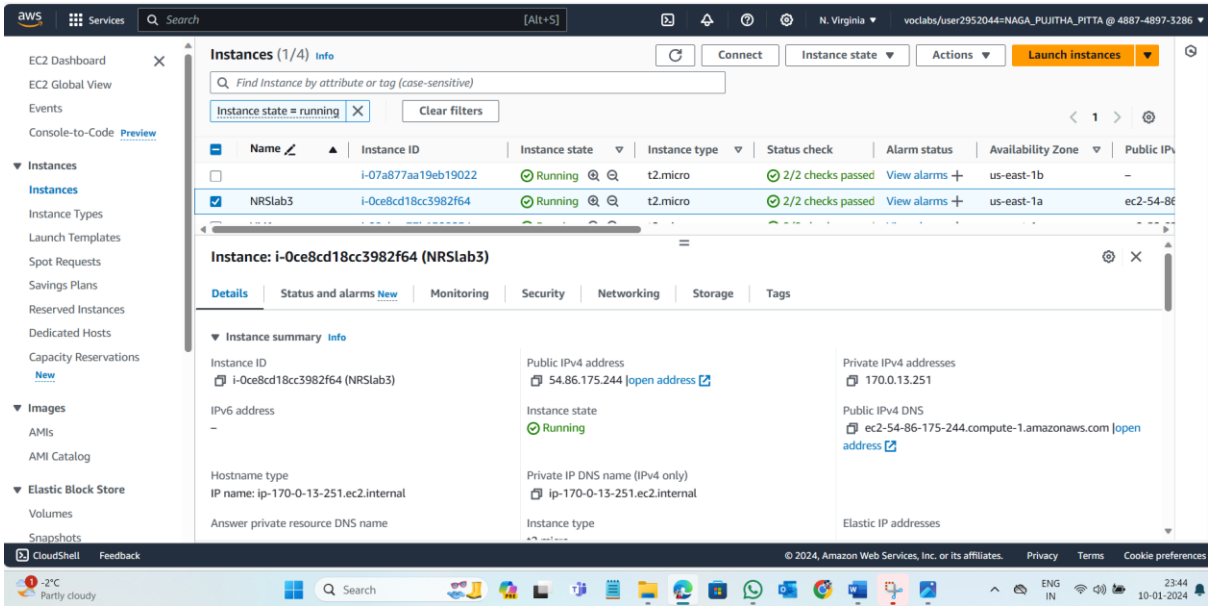
EC2 instance has been connected.



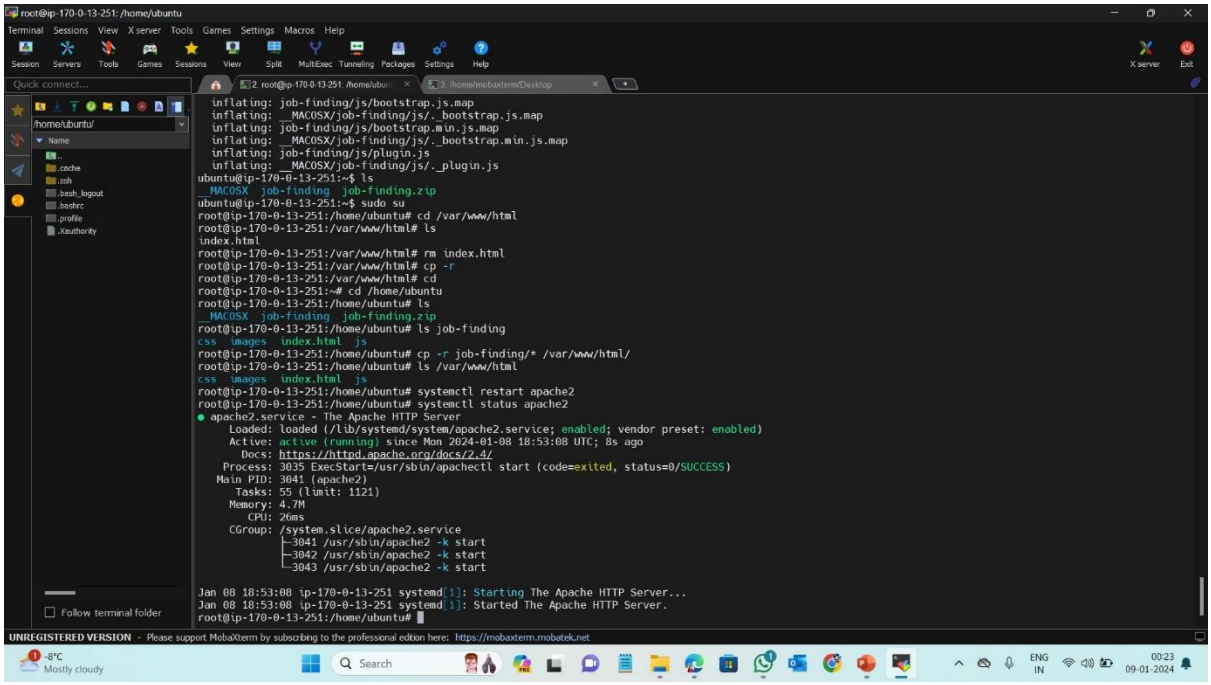
After successfully installing the job-finding application, it became essential to build a temporary directory on the present server. As a result, the job application had to be extracted into this specified temporary location.



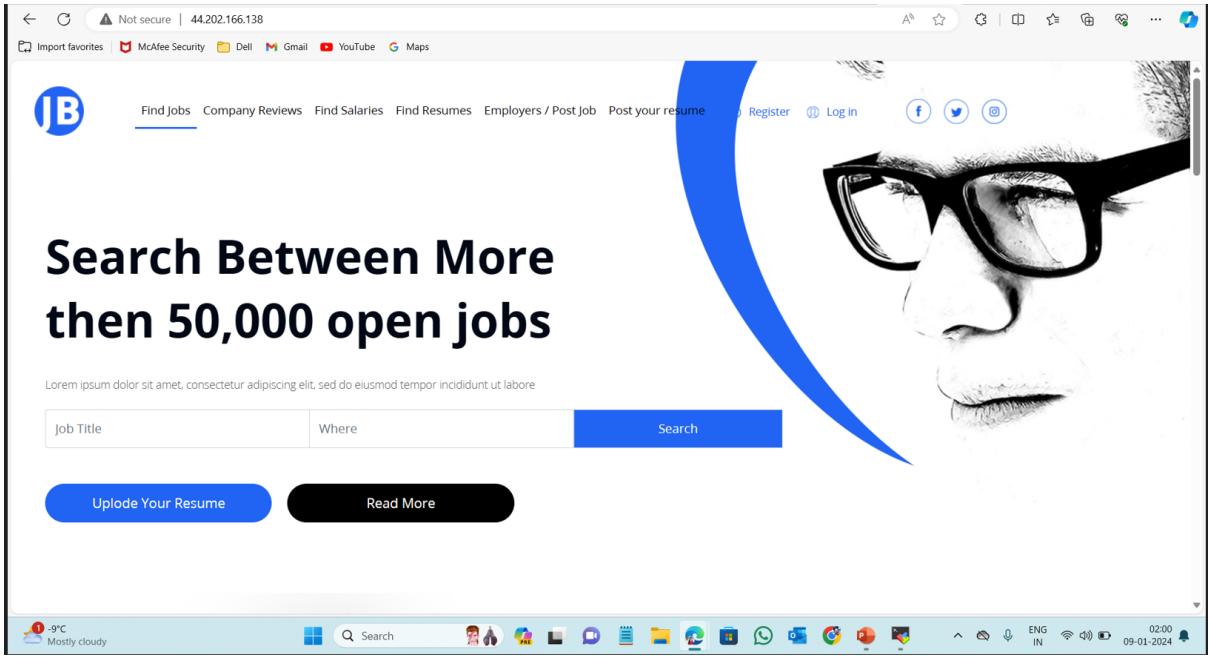




If the web application was not connected to the server, it was essential to check the status of the httpd server using the command systemctl status httpd. If the server became inactive, it had to be restarted with the command systemctl start httpd.

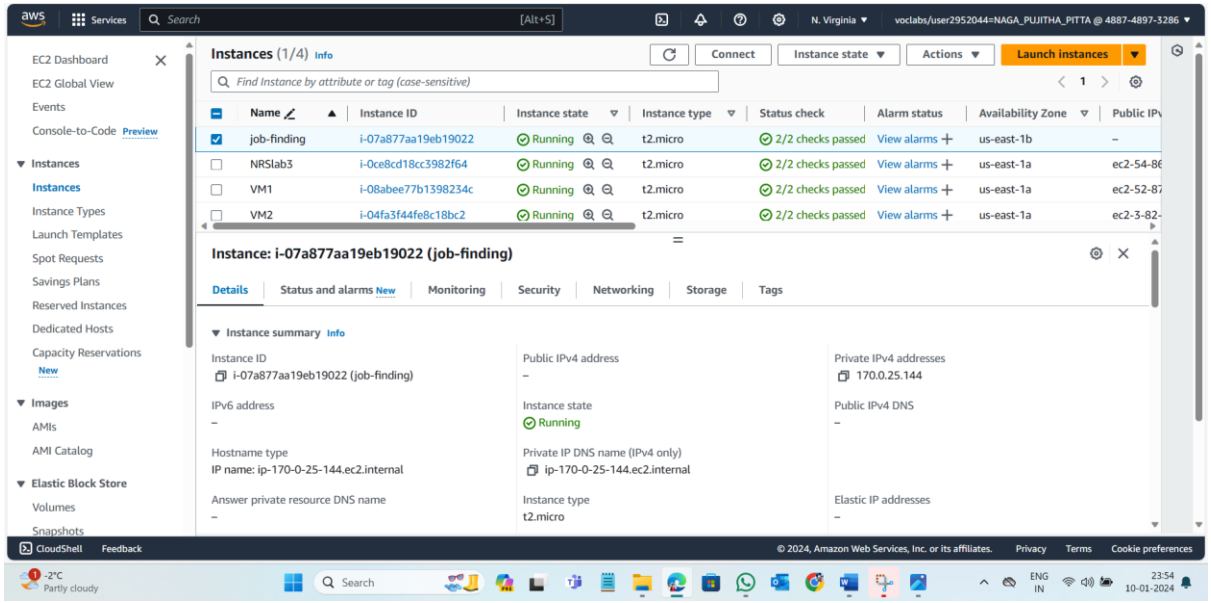


Consequently, the web application was effectively hosted on the existing server, accessible through the public IP address.

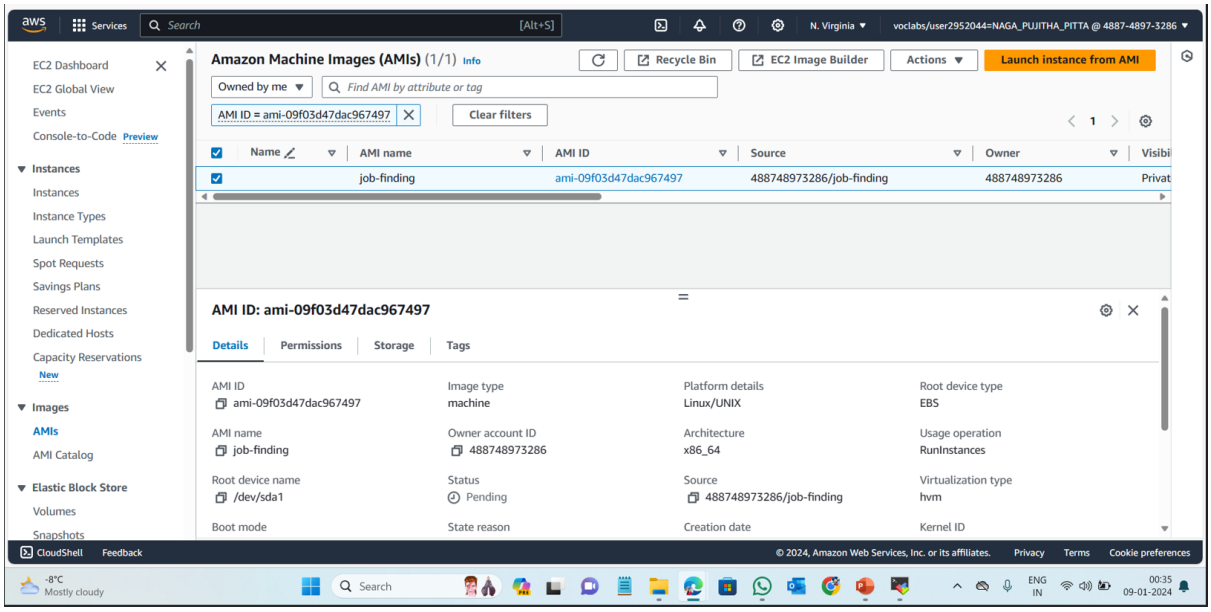


The "job-finding" web application has been successfully hosted on the web server. However, the present web server instance does not support auto-scaling. The server load may increase as user traffic increases, thus affecting performance. Autoscaling for the server and the assignment of a load balancer are required to handle this problem.

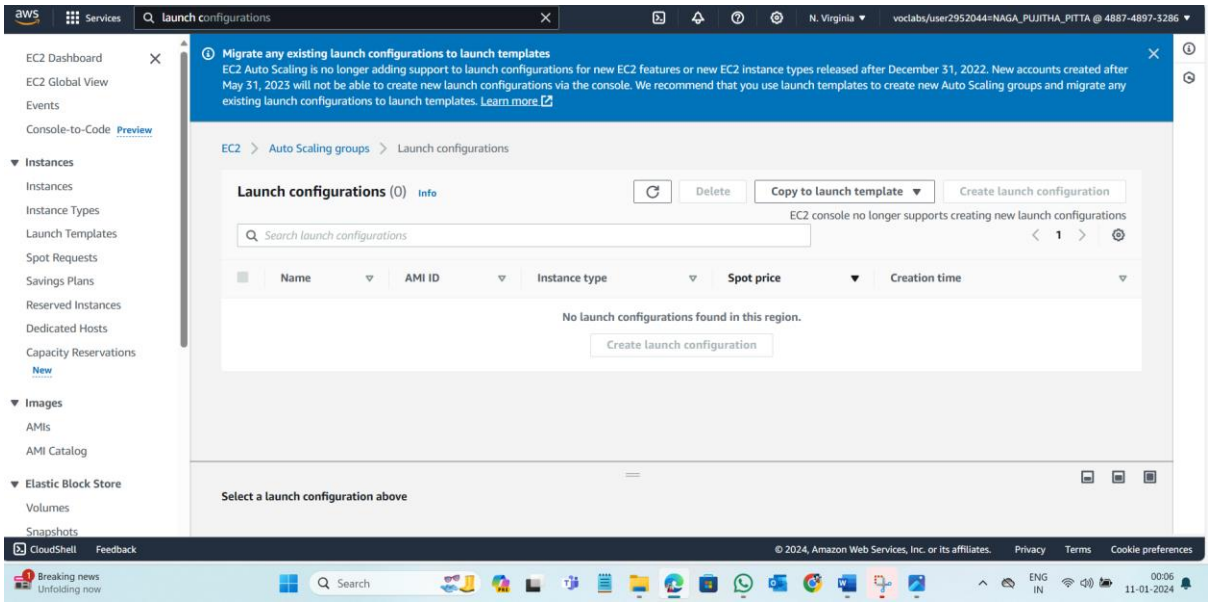
Following the deployment of the web application, the next step is to create a launch configuration to allow autoscaling groups to function. It is critical to duplicate the web server configurations before establishing the launch configuration. Creating an image that duplicates all configurations similar to the root web server is required.



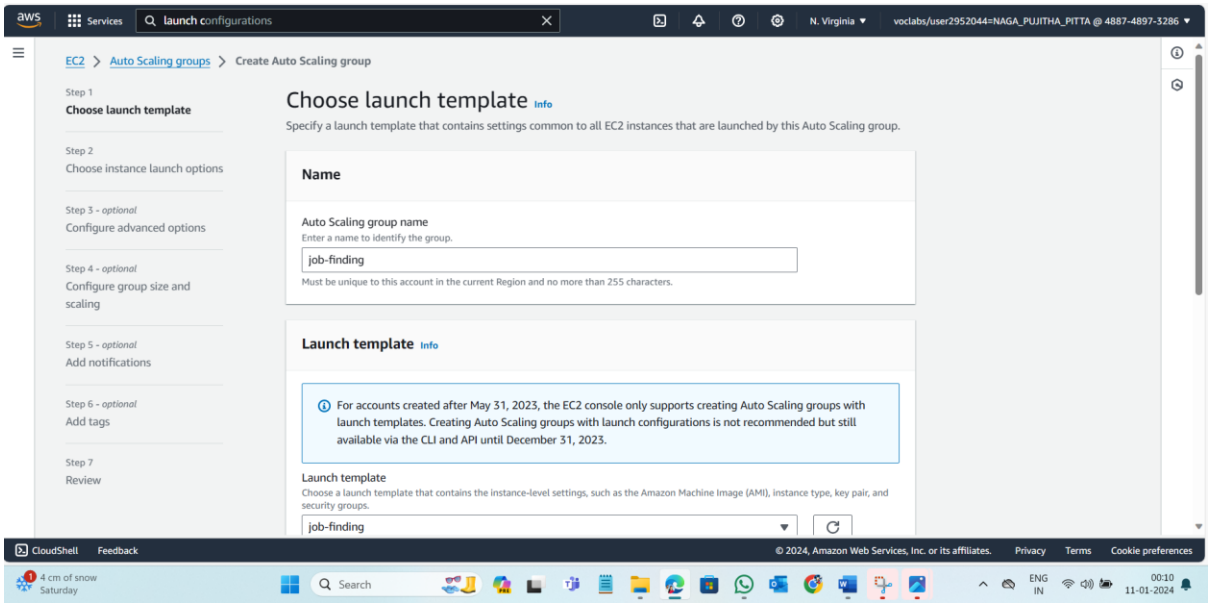
The image was successfully produced.



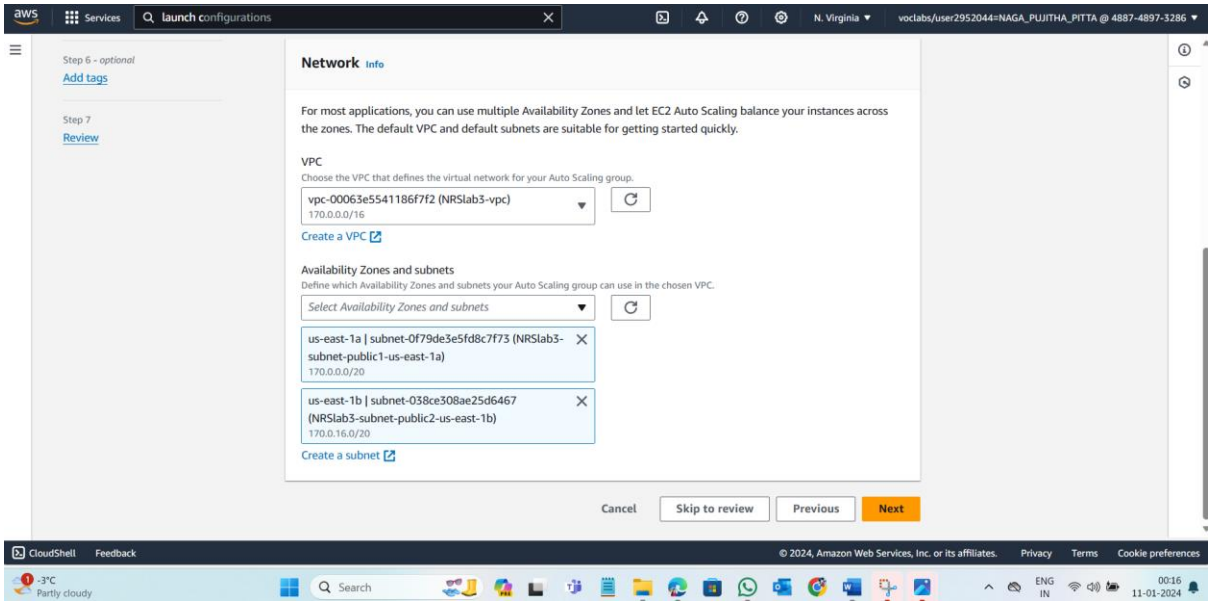
We have to build the launch configuration with the name “job-finding”.



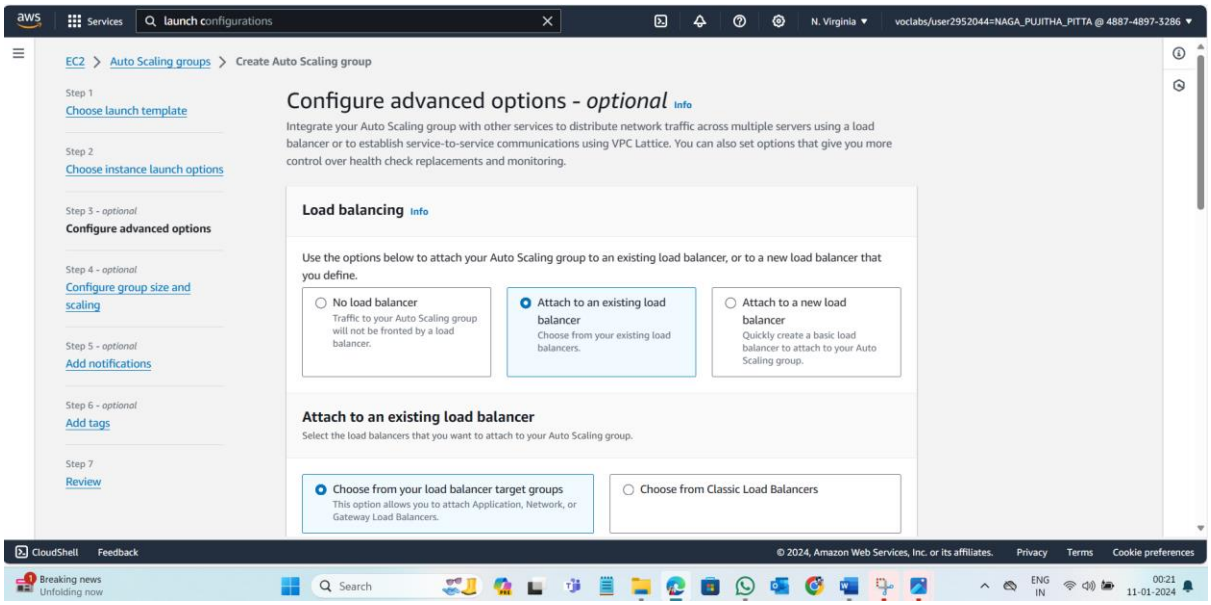
After successfully hosting the website, the following step is to configure autoscaling. However, before starting autoscaling, you must first create an autoscaling group.

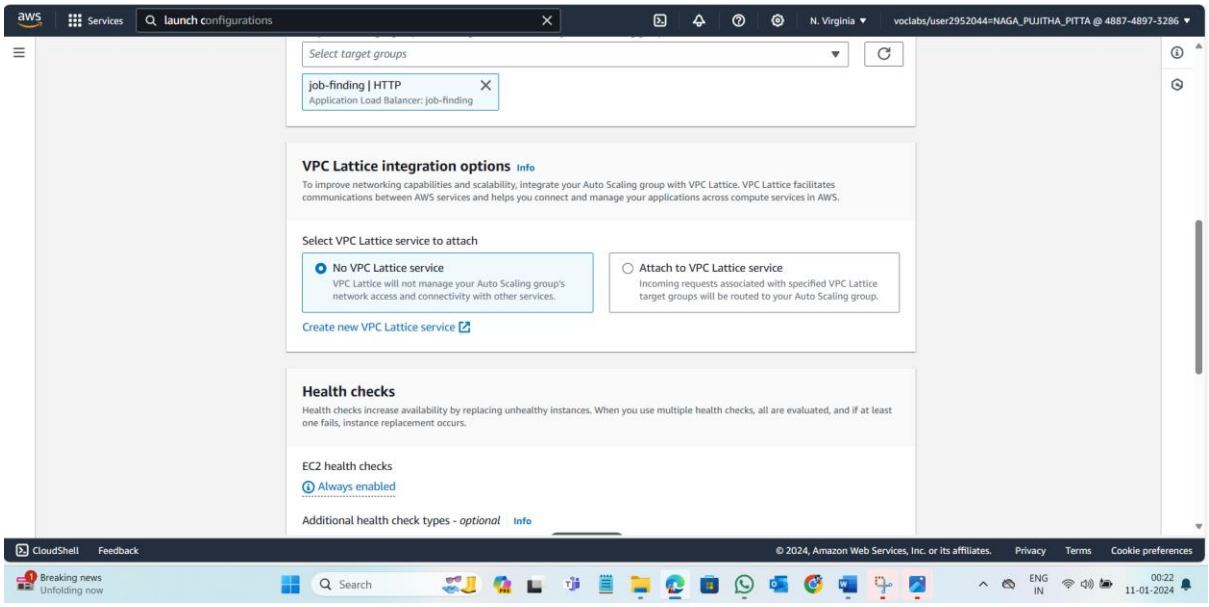


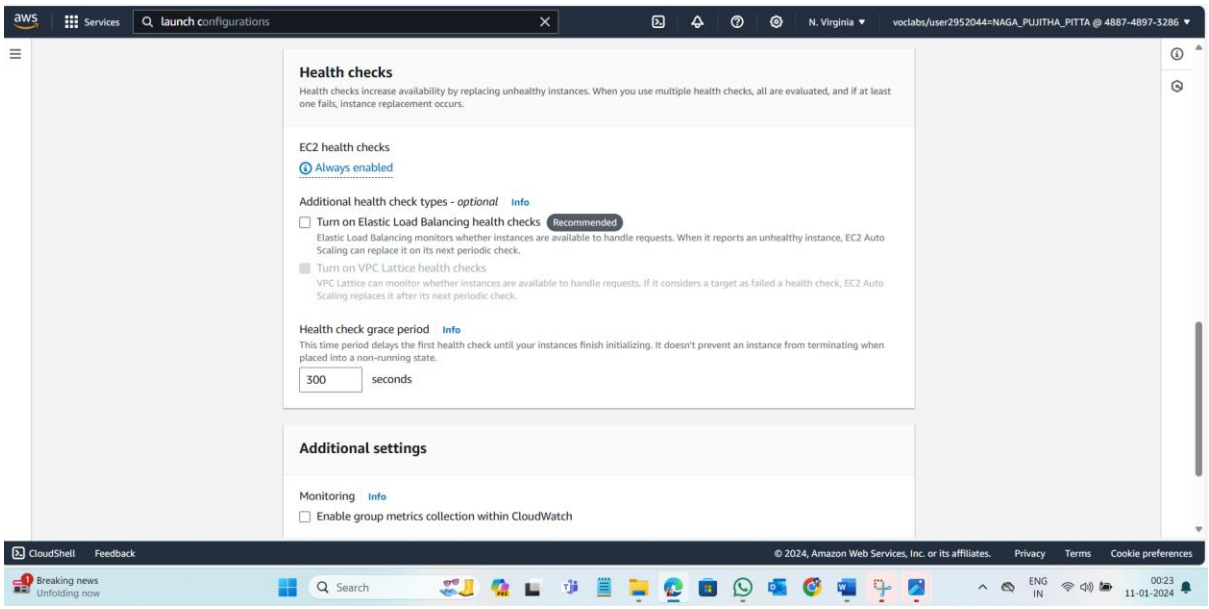
Two specific availability zones, namely us-east-1a and us-east-1b, were employed.



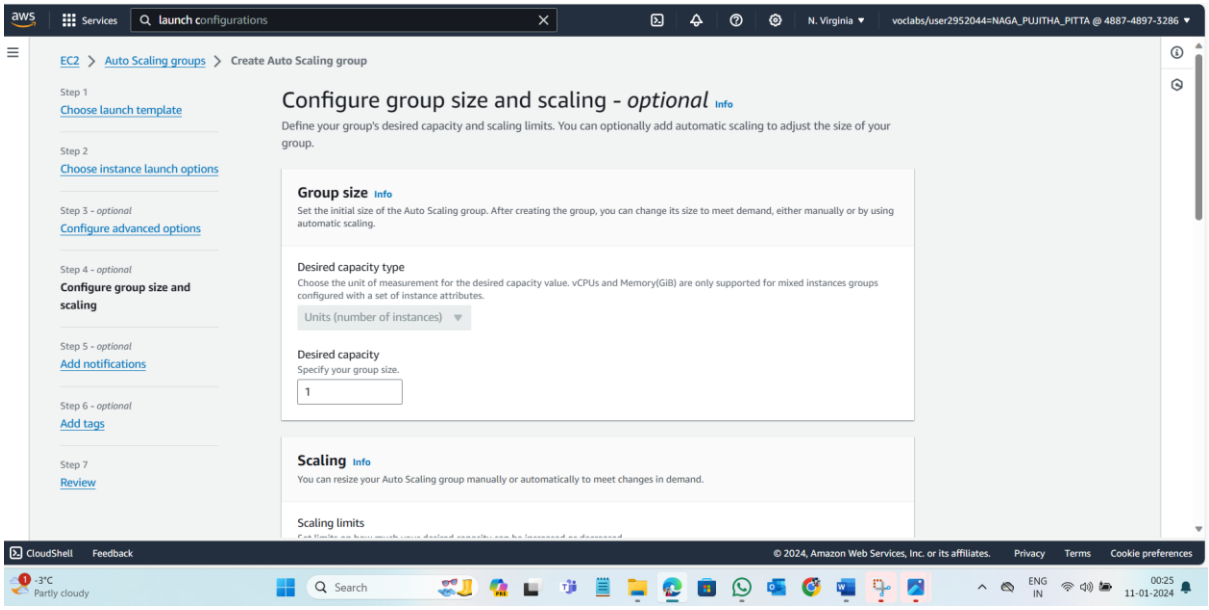
It is critical to create a connection at this point, with the load balancer already built. The following step is to add the current load balancer and associate it with an autoscaling group. As a result, the option to add an existing load balancer should be chosen, and a name, such as "job-finding," should be given.

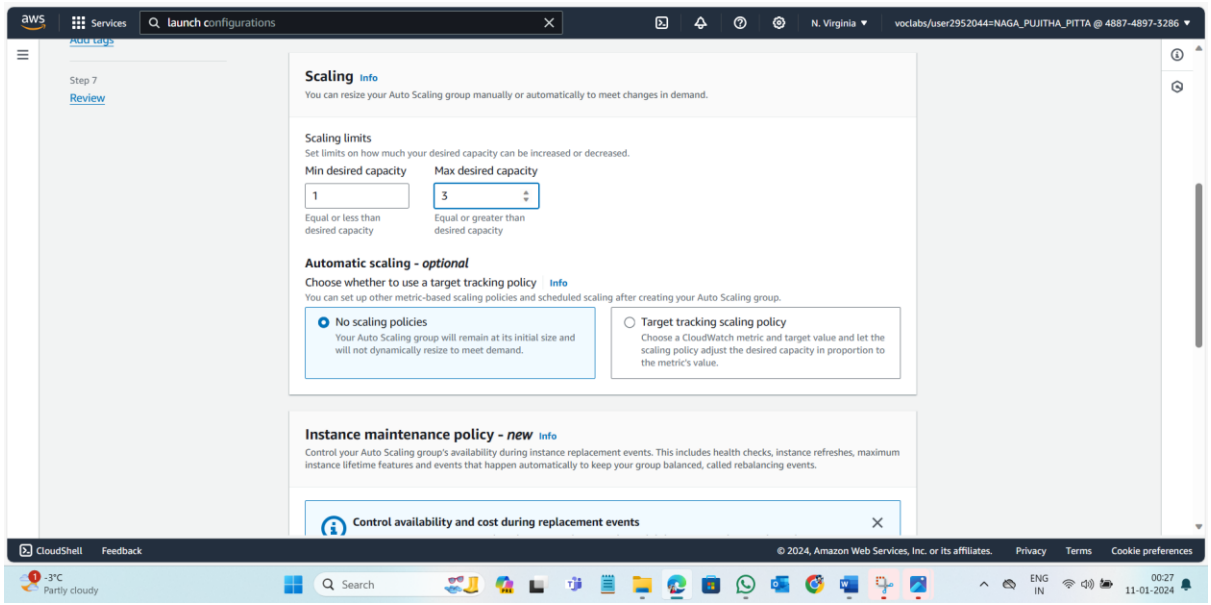


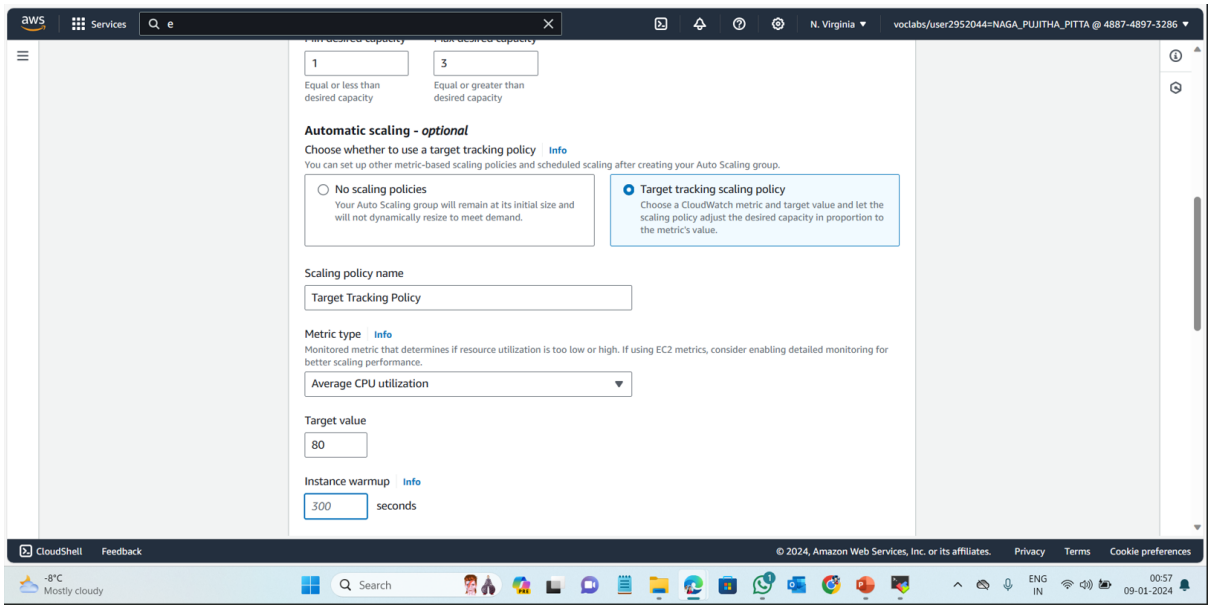




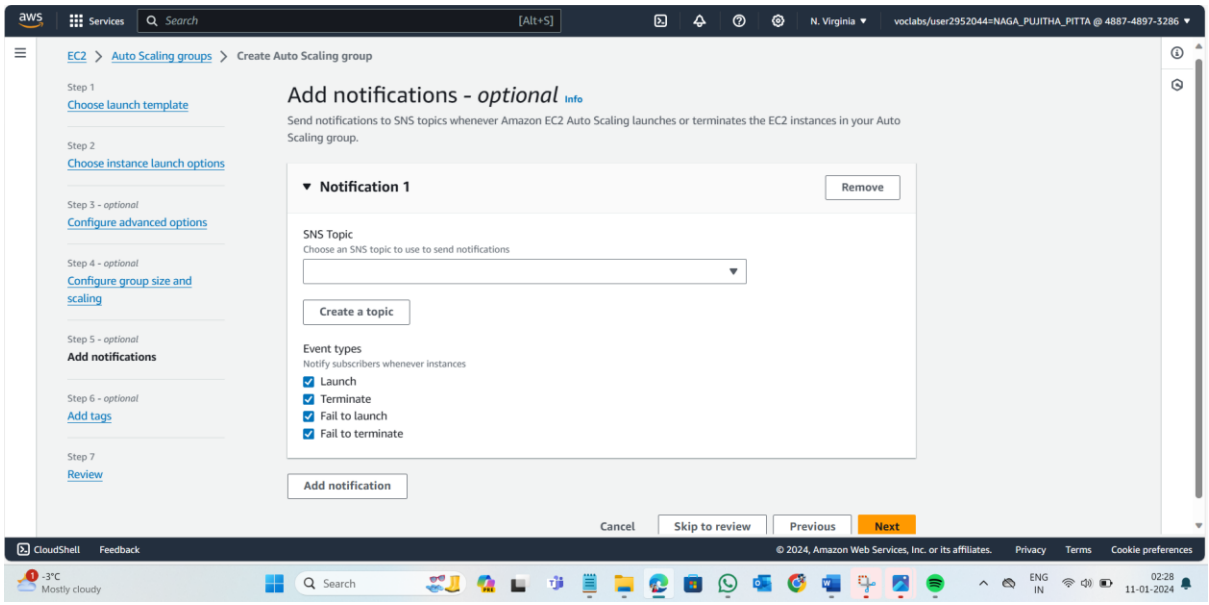
It is critical in this scenario to provide the group size for the scaling groups, with the targeted capacity set to 1, the minimum capacity set to 1, and the maximum capacity set to 3.



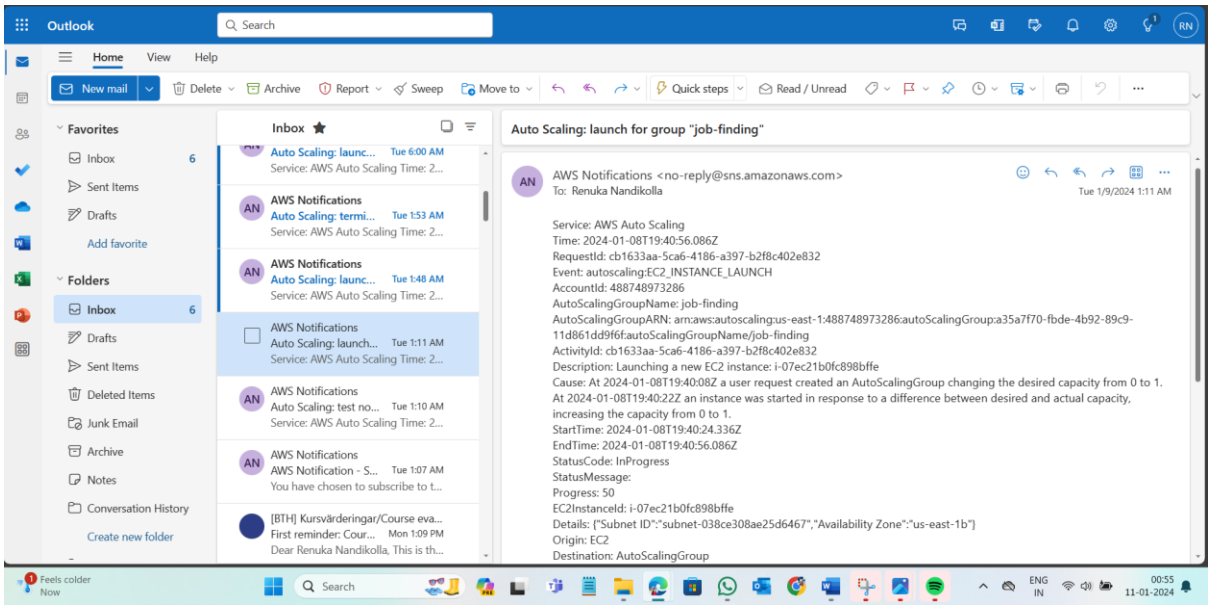

launch configurations
N. Virginia
voclabs/user2952044=NAGA_PUJITHA_PITTA @ 4887-4897-3286
Step 7
Review
Scaling Info
You can resize your Auto Scaling group manually or automatically to meet changes in demand.
Scaling limits
Set limits on how much your desired capacity can be increased or decreased.
Min desired capacity
Max desired capacity
1
3
Equal or less than desired capacity
Equal or greater than desired capacity
Automatic scaling - optional
Choose whether to use a target tracking policy Info
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.
No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.
Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.
Instance maintenance policy - new Info
Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.
Control availability and cost during replacement events
CloudShell
Feedback
© 2024, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
-3°C
Partly cloudy
Search
ENG
IN
00:27
11-01-2024


e
N. Virginia
voclabs/user2952044=NAGA_PUJITHA_PITTA @ 4887-4897-3286
1
3
Equal or less than desired capacity
Equal or greater than desired capacity
Automatic scaling - optional
Choose whether to use a target tracking policy Info
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.
No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.
Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.
Scaling policy name
Target Tracking Policy
Metric type Info
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.
Average CPU utilization
Target value
80
Instance warmup Info
300
seconds
CloudShell
Feedback
© 2024, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
-8°C
Mostly cloudy
Search
ENG
IN
00:57
09-01-2024

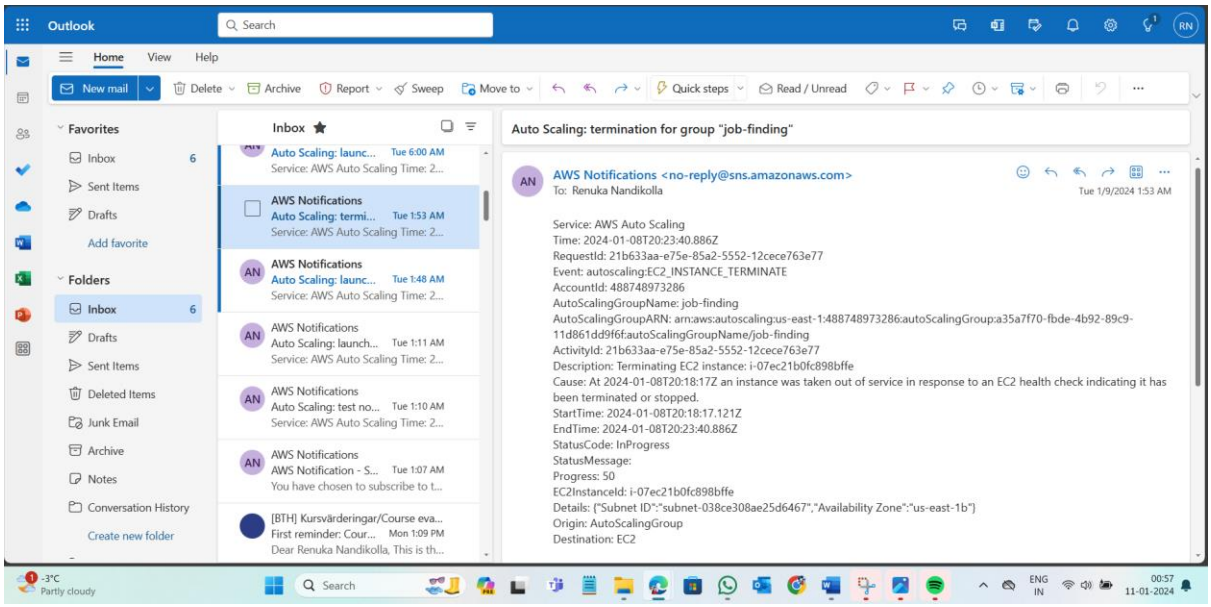
At this point, a notification mechanism is in place. When the CPU utilization exceeds 80%, a message is sent and the server is promptly moved to another server.



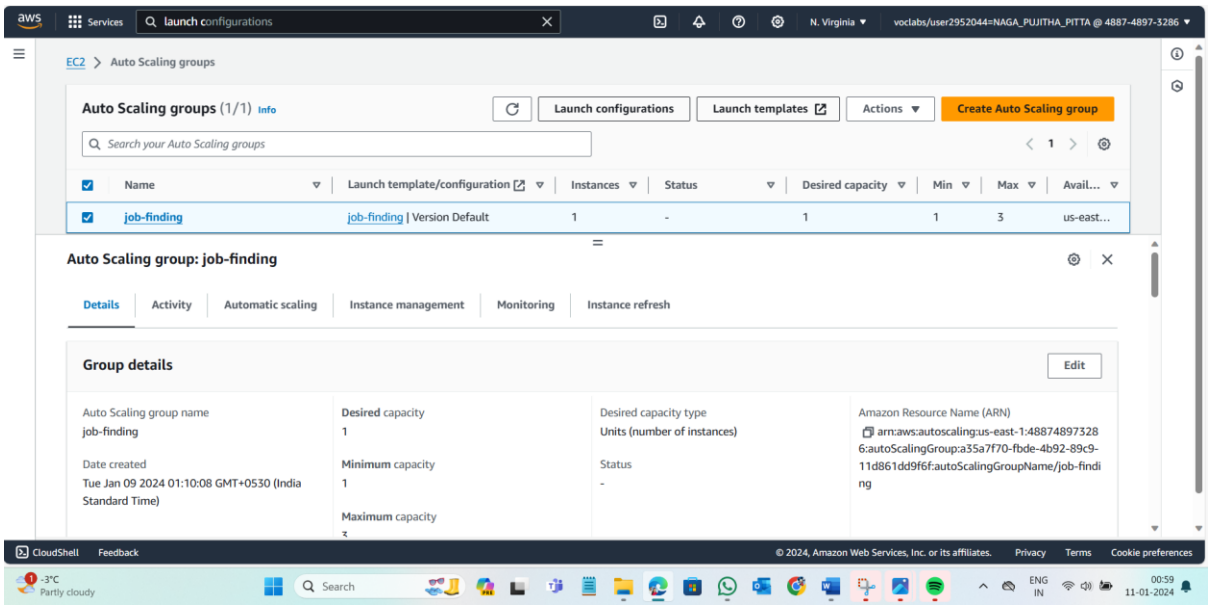
AWS Auto Scaling was initiated, and a notice was sent to the email address.



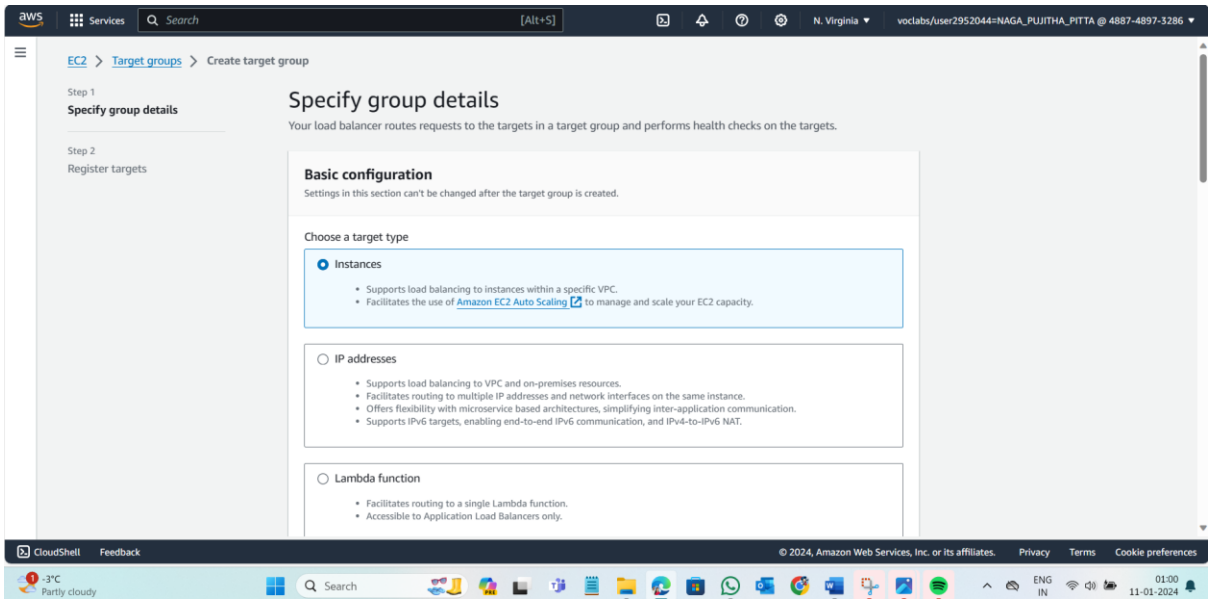
Furthermore, the AWS Auto Scaling instance was taken from operation as a result of an EC2 health check, signaling termination or stop.

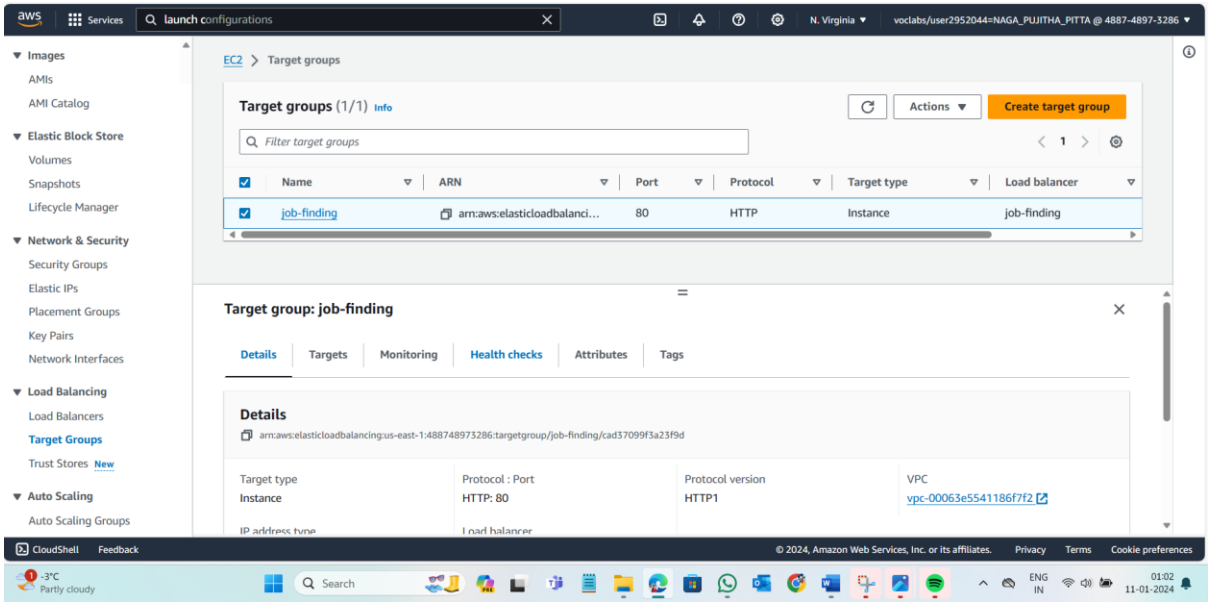


Auto Scaling group with the name job-finding has been successfully built.



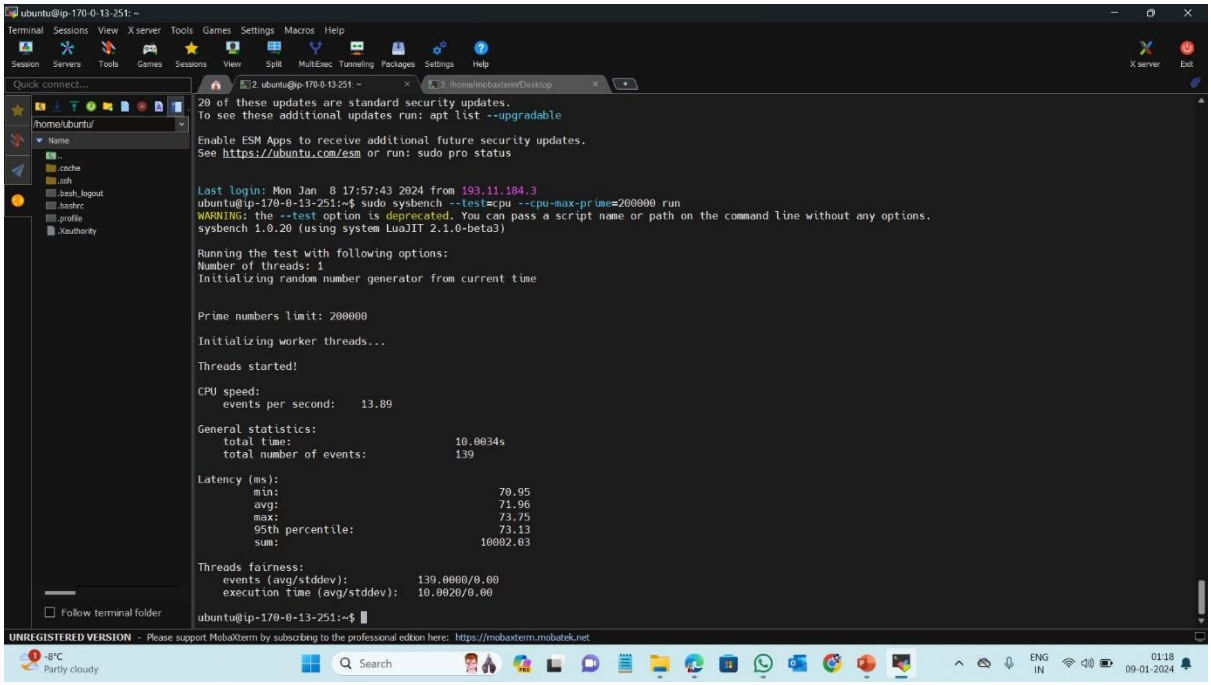
First, a target group must be created, which entails choosing a target type (in this case, instances) and giving the target group a name. The next step is to pick an existing VPC for the target group. The last step is to select HTTP1 as the protocol version for the target group.



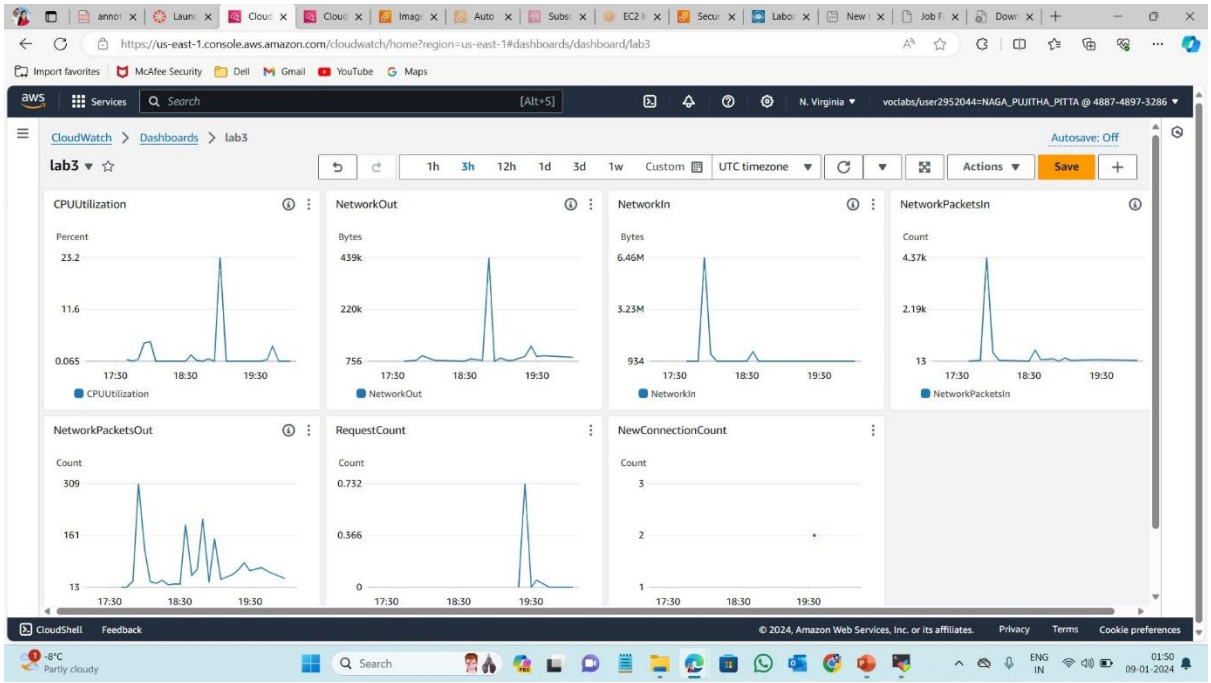


❖ Testing:

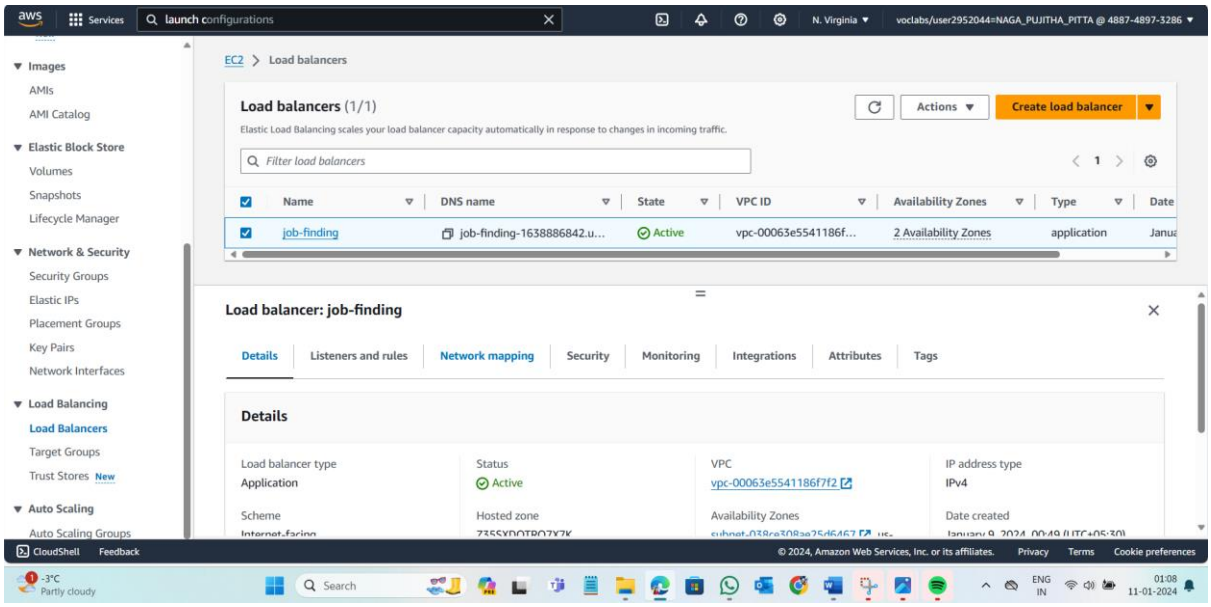
Conducting an Apache Bench stress test.



❖ Monitoring Metrics:



The subsequent step is to register the instance with the registered targets when a target group has been successfully created. The public DNS name can be used for the validation. The application will be hosted using the public DNS name if the validation is complete.



The Website has been successfully hosted.

